8.033 Fall 2025: Solution Set 9

1

(a) When the astronaut is closest to the earth as measured in the earth frame, the radio waves are transmitted to the spaceship along a direction that is perpendicular to the trajectory of the spaceship. This is the case of transverse Doppler shift discussed in class. Using the formula given in the lecture notes, we get

$$f_o = \gamma f_s = \frac{1}{\sqrt{1-v^2}} f_s = \frac{1}{\sqrt{1-(1/2)^2}} f_s = 808kHz. \tag{1}$$

(b) We can solve this problem by staying in the earth's frame, knowing that the time on earth t and the proper time of the astronaut τ are simply related by $t = \gamma\tau$. Suppose the astronaut was at her closet approach to the earth at time $t = 0$. Then at a later time $t > 0$, the astronaut has traveled a distance vt, measured in the earth frame (see Fig [1]), so the angle between the radio waves' trajectory and the spaceship' velocity are given by

$$\cos(\theta_s) = \frac{vt}{\sqrt{(vt)^2 + L^2}}. \tag{2}$$

We can use the formula of relativistic Doppler shift Eq. (7.7) from the lecture notes to find out the frequency of the radio waves received by the astronaut

$$f_o(t) = \gamma f_s(1 - v\cos\theta_s) = \gamma f_s(1 - \frac{v^2 t}{\sqrt{(vt)^2 + L^2}}). \tag{3}$$

Now using $t = \gamma\tau$, we get

$$f_o(\tau) = \gamma f_s(1 - \frac{v^2\gamma\tau}{\sqrt{(v\gamma\tau)^2 + L^2}}). \tag{4}$$

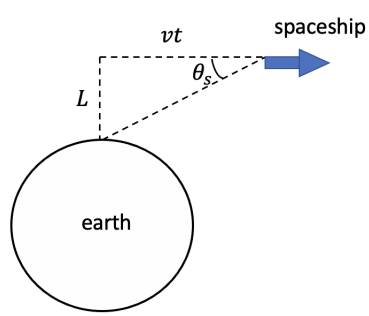

Figure 1: The spaceship seen in the earth frame at time t.

2

Suppose it is possible for an electron and a photon to collide and result in a single electron. Then the conservation of 4-momentum requires

$$p^\mu + q^\mu = (p')^\mu,$$

where p^μ, q^μ, and $(p')^\mu$ represent the 4-momenta of the electron and photon before collision, and the electron after collision, respectively. Since the 4-vectors on the left and right-hand side of Eq. (21) are equal, the invariant product with themselves are also equal, so we have

$$\begin{aligned}(p^\mu + q^\mu)(p_\mu + q_\mu) &= (p')^\mu (p')_\mu \\ p^\mu p_\mu + p^\mu q_\mu + p_\mu q^\mu + q^\mu q_\mu &= (p')^\mu (p')_\mu.\end{aligned}$$

Since the invariant product is invariant under coordinate transformations, we can evaluate each term in Eq. (23) in a reference frame that is most convenient for our calculation. For example, in the rest frame of the electron, its 4-momentum is simply

$$p^\mu \to \begin{bmatrix} m_e \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

(where we have set $c = 1$) so we get $p^\mu p_\mu = -m_e^2$. Similarly, we can choose our reference frame such that the photon propagates along the x-axis, so

$$q^\mu \to \begin{bmatrix} E_\gamma \\ E_\gamma \\ 0 \\ 0 \end{bmatrix},$$

and we get $q^\mu q_\mu = 0$. Therefore, Eq. (23) becomes

$$-m_e^2 + 2p^\mu q_\mu + 0 = -m_e^2,$$

or equivalently

$$p^\mu q_\mu = 0.$$

We can compute this in the rest frame of the electron before the collision, where

$$p^\mu \to \begin{bmatrix} m_e \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

and

$$q_\mu \to \begin{bmatrix} -E_\gamma \\ p^x \\ p^y \\ p^z \end{bmatrix}.$$

Then Eq. (27) becomes $-m_e E_\gamma = 0$, indicating $E_\gamma = 0$. Therefore, there is no photon that can collide with an electron and only results in an electron.

3

By definition, the 4-momentum of a particle is $p^\mu = mU^\mu$, and recall from pset 5 problem 5 that the 4-velocity U^μ has components

$$\begin{bmatrix} U^0 \\ U^1 \\ U^2 \\ U^3 \end{bmatrix} = \gamma \begin{bmatrix} 1 \\ V^1 \\ V^2 \\ V^3 \end{bmatrix},$$

so

$$p^\mu \to m\gamma \begin{bmatrix} 1 \\ V^1 \\ V^2 \\ V^3 \end{bmatrix}.$$

Since the particle in our case is moving along the x-axis only, the above becomes

$$p^\mu \to m\gamma \begin{bmatrix} 1 \\ V \\ 0 \\ 0 \end{bmatrix}.$$

Since $\gamma = 1/\sqrt{1-V^2}$, we can write V in terms of γ and get

$$V = (1-\gamma^{-2})^{1/2},$$

so the x component of the 4-momentum becomes

$$p^x = m\gamma V = m\gamma(1-\gamma^{-2})^{1/2} = (\gamma^2-1)^{1/2}m.$$

In other words, we have the 4-momentum to be

$$p^\mu \to m \begin{bmatrix} \gamma \\ (\gamma^2-1)^{1/2} \\ 0 \\ 0 \end{bmatrix}.$$

4

By the conservation of 4-momentum, we have

$$(p_1)^\mu = (p_2)^\mu + q^\mu, \tag{36}$$

where $(p_1)^\mu$, $(p_2)^\mu$, q^μ represent the 4-momenta of the particle with rest mass m_1, the particle with rest mass m_2, and the photon, respectively. We can compute the invariant product of both sides of Eq. (36)

$$\begin{aligned}(p_1)^\mu(p_1)_\mu &= [(p_2)^\mu + q^\mu][(p_2)_\mu + q_\mu] \\ &= (p_2)^\mu(p_2)_\mu + 2(p_2)^\mu q_\mu + q^\mu q_\mu.\end{aligned} \tag{37}$$

By the same reasoning as in problem 3, we easily conclude that $(p_1)^\mu(p_1)_\mu = -m_1^2$, $(p_2)^\mu(p_2)_\mu = -m_2^2$, and $q^\mu q_\mu = 0$. We evaluate $(p_2)^\mu q_\mu$ in the rest frame of the particle with rest mass m_2 and we get $(p_2)^\mu q_\mu = -m_2 E_\gamma$, where E_γ is the photon energy in the rest frame of the produced particle. Eq. (37) now becomes

$$-m_1^2 = -m_2^2 - 2m_2 E_\gamma, \tag{38}$$

so we get

$$E_\gamma = \frac{m_1^2 - m_2^2}{2m_2}c^2. \tag{39}$$

5

By the conservation 4-momentum, we have

$$(p_p)^\mu + q^\mu = (p_\pi)^\mu + (p_n)^\mu, \tag{40}$$

where $(p_p)^\mu$, q^μ, $(p_\pi)^\mu$, and $(p_n)^\mu$ represent the momenta of the proton, photon, π meson, and the neutron, respectively. We can compute the invariant product of both sides and get

$$[(p_p)^\mu + q^\mu][(p_p)_\mu + q_\mu] = [(p_\pi)^\mu + (p_n)^\mu][(p_\pi)_\mu + (p_n)_\mu] \tag{41}$$

First, let's examine the right-hand side of Eq. (41). It is important to realize that the reaction

$$p + \gamma \to \pi^+ + n$$

can only occur if the proton is energetic enough. In the limiting case of the lowest possible proton energy for the reaction to occur, we will have the π meson and neutron produced at rest in their center of momentum frame. In that frame, we have

$$(p_\pi)^\mu + (p_n)^\mu \to \begin{bmatrix} m_\pi + m_n \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{42}$$

so we know that $[(p_\pi)^\mu + (p_n)^\mu][(p_\pi)_\mu + (p_n)_\mu] = -(m_\pi + m_n)^2$.

Now, let's examine the left-hand side of Eq. (41)

$$[(p_p)^\mu + q^\mu][(p_p)_\mu + q_\mu] = -m_p^2 + 2(p_p)^\mu q_\mu. \tag{43}$$

In the limiting case of the lowest proton energy, the proton must collide head on with the photon in the lab frame (where the photon energy is measured to be $6 \cdot 10^{-4} eV$ as given by the problem), because otherwise the proton would carry some "extra" energy from momentum that is perpendicular to the photon's momentum. Align the x-axis of the lab frame such that the photon is traveling in the -x direction, then in the lab frame

$$q^\mu \to \begin{bmatrix} E_\gamma \\ -E_\gamma \\ 0 \\ 0 \end{bmatrix}, \tag{44}$$

and

$$p^\mu \to \begin{bmatrix} E_p \\ E_p \\ 0 \\ 0 \end{bmatrix}, \tag{45}$$

where E_p is the proton energy in the lab frame. We have approximated that all of the proton energy comes from its momentum, which is valid in this case since the proton is

highly relativistic ($\gamma \gg 1$). Then we get $(p_p)^\mu q_\mu = -2E_p E_\gamma$. Putting everything together, Eq. (41) becomes

$$m_p^2 + 4E_p E_\gamma = m_\pi^2 + m_n^2 + 2m_\pi m_n. \tag{46}$$

Use $m_n = m_p$ and solve for the proton energy, we get

$$E_p = \frac{2m_\pi m_p + m_\pi^2}{4E_\gamma} c^4, \tag{47}$$

where a factor of c^4 has been inserted for the correct dimension. Plugging in numerical values, we get

$$E_p = 1.2 \cdot 10^{20} eV. \tag{48}$$